

Docker Tutorial – Apache2 Web Service

1. What is Docker?

Docker is a platform that packages applications and their dependencies into containers. Containers are lightweight, portable, and ensure consistency across different environments.

Benefits of Docker:

- Portability
- Isolation
- Scalability
- Faster deployment

2. Creating a Dockerfile for Apache2

Directory structure:

myapp/

├─ Dockerfile

└─ index.html

Contents of index.html:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Welcome to Apache2 in Docker</title>
```

```
</head>
```

```
<body>
```

```
<h1>Hello from Dockerized Apache2!</h1>
```

```
</body>
```

```
</html>
```

Contents of Dockerfile:

FROM ubuntu

RUN apt update && apt upgrade -y && apt install apache2 -y

WORKDIR /srv/myapp

COPY ./index.html /var/www/html/index.html

ENTRYPOINT ["apachectl", "-D", "FOREGROUND"]

3. Building and Running the Docker Image

Build the image:

docker build -t myapp .

Run the container:

Shell

docker run -d -p 80:80 myapp

Visit the cloud instance IP in your browser to see the page.

4. Pushing to Docker Hub (Not required)

Step 1: Login to Docker Hub

docker login

Step 2: Tag the image

docker tag myapp username/myapp:latest

Step 3: Push the image

docker push username/myapp:latest

Step 4: Pull the image on another machine

```
docker pull username/myapp:latest
```

```
docker run -d -p 80:80 username/myapp:latest
```